

DataCORE

Python, Pandas & Your Team Dataset

June 30th, 2026

Afternoon: DataCORE

1. **Python Fundamentals — Variables & Data Types**
2. **Pandas — Loading & Describing Your Team Dataset**
3. **Scientific Storytelling — Variable Definitions**

The same variables you defined on paper this morning — now we store them in code.

1:00 – 1:45 · Python Fundamentals

Variables store information

- A **variable** in research stores information — so does a **variable in Python**.
- This morning: **IV** = hours_of_sleep, **DV** = test_score.
- **PBL question:** how do we store these in Python?
→

Big idea: the **name** on the left holds the **value** on the right. Use it again just by typing its name.

Python	R
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```
hours_of_sleep = 7.5  # our IV: how long a student slept
test_score     = 88   # our DV: the outcome we measure

print(hours_of_sleep, test_score)  # use them by name
```

7.5 88

String vs. integer vs. float

Every value has a **type**. Python tracks it for you — `type()` reveals it.

Python	R
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```
student_name = "Maria"  # string -> text in quotes
grade_level  = 10       # integer -> a whole number
hours_of_sleep = 7.5    # float -> a number with a decimal

print(type(student_name), type(grade_level), type(hours_of_sleep))
```

```
<class 'str'> <class 'int'> <class 'float'>
```

💡 Type matters: you can do **math** on numbers, but not on text.

Lists store many values

One variable can hold a **whole column** of values — like every student's sleep.

Python

R

```
sleep_hours = [7.5, 6.0, 8.0, 5.5]  # a list of values
```

```
print(len(sleep_hours))            # how many values?
```

4

```
print(sum(sleep_hours) / len(sleep_hours))  # the average
```

6.75

 A dataset is just **many lists side by side** — that's what Pandas gives us next.

1:45 – 2:45 · Pandas

Pick your team dataset

Each team works with a **real dataset** from this pre-curated list:



CDC

Health & disease



FRED

Economics & jobs



NOAA

Weather & climate



WHO

Global health



Sports-Reference

Teams, players, stats

 The examples below use a sample `study_habits.csv` — swap in **your** file.

Step 1 · Load the file with `pd.read_csv()`

Python R

```
import pandas as pd
df = pd.read_csv("study_habits.csv")  # load the CSV into a DataFrame
print(df.head())                    # peek at the first 5 rows
```

	student_id	grade_level	study_method	hours_of_sleep	test_score
0	1	9th	solo	7.5	82.0
1	2	9th	group	6.0	74.0
2	3	9th	tutor	8.0	88.0
3	4	10th	solo	5.5	68.0
4	5	10th	group	7.0	79.0

 Each **row** is an observation; each **column** is a variable.

Step 2 · df.info() – columns & data types

Python

R

```
df.info()    # column names, non-null counts, and data types
```

```
<class 'pandas.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 5 columns):
 #   Column          Non-Null Count  Dtype  
---  -
 0   student_id      20 non-null    int64  
 1   grade_level     20 non-null    str     
 2   study_method    20 non-null    str     
 3   hours_of_sleep  19 non-null    float64 
 4   test_score      19 non-null    float64 
dtypes: float64(2), int64(1), str(2)
memory usage: 928.0 bytes
```

 Which columns are **numbers**? Which are **text/categories**?

! Step 3 · `df.isnull().sum()` – spot missing data

Python R

```
df.isnull().sum()    # count missing values in each column
```

```
student_id      0  
grade_level     0  
study_method    0  
hours_of_sleep  1  
test_score      1  
dtype: int64
```

Missing data is normal. Knowing *where* it hides protects you from wrong conclusions later.

Step 4 · value_counts() – frequency of categories

Python R

```
df["study_method"].value_counts()    # how often each category appears
```

```
study_method
solo        7
group       7
tutor       6
Name: count, dtype: int64
```

 Great for **categorical** variables — grade level, method, region, team.

AI assist – then verify

1. Paste your **column names** into ChatGPT.
 2. Ask: *"What does each column likely measure?"*
 3. **Verify** against the actual data with `info()` and `value_counts()`.
 4. **Discuss discrepancies** as a team.
-  Did the AI misread any column? Note it on your **Mission Sheet**.

AI is helpful – but not always right. It *guesses* from names; only the **data** tells the truth. When they disagree, trust what you can check.

2:45 – 3:00 · Scientific Storytelling

Define your IV & DV in one sentence each

As a team, write a **one-sentence definition** of your **IV** and **DV** — and **cite one source**.

Class checks:

- Is it **specific**?
- Is it **citable**?

Practice in-text citation: **(Author, Year)**.

 Read aloud. Is every definition **specific** and **citable**?

*"Our independent variable, **hours of sleep**, is the nightly time a student sleeps before a test, which prior work links to memory consolidation (Walker, 2017)."*

*"Our dependent variable, **test score**, is the percentage correct on a standardized exam (College Board, 2022)."*